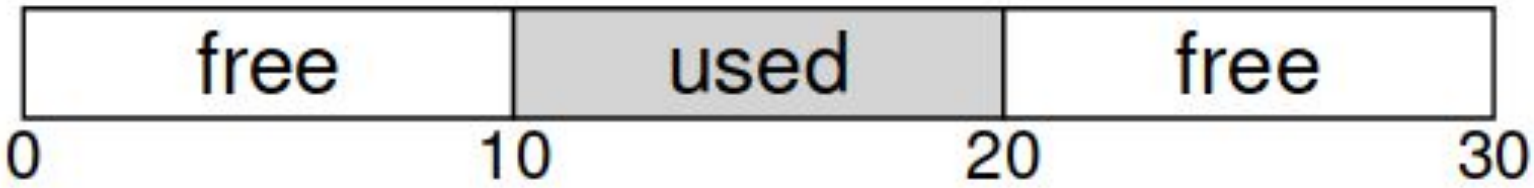
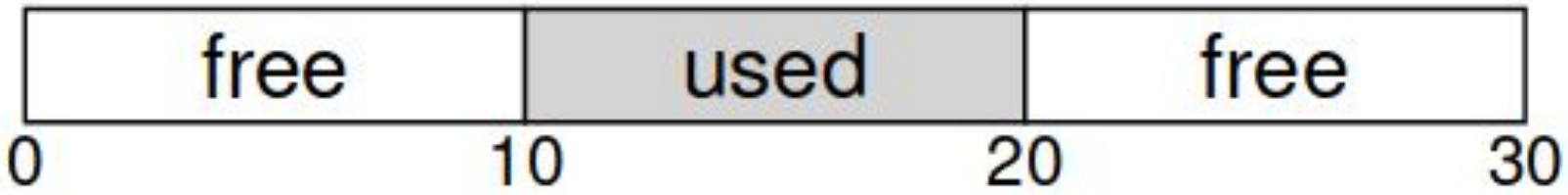
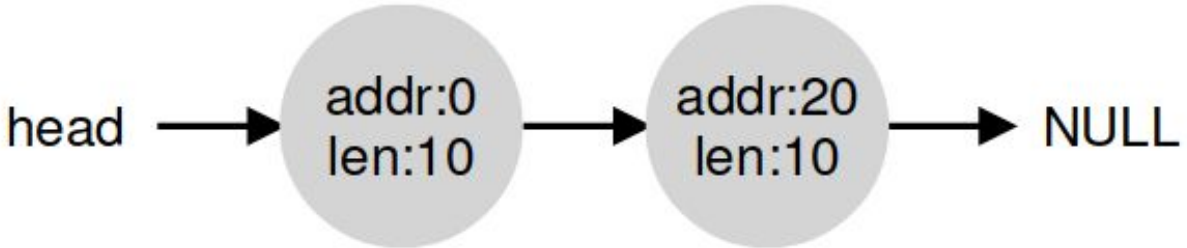
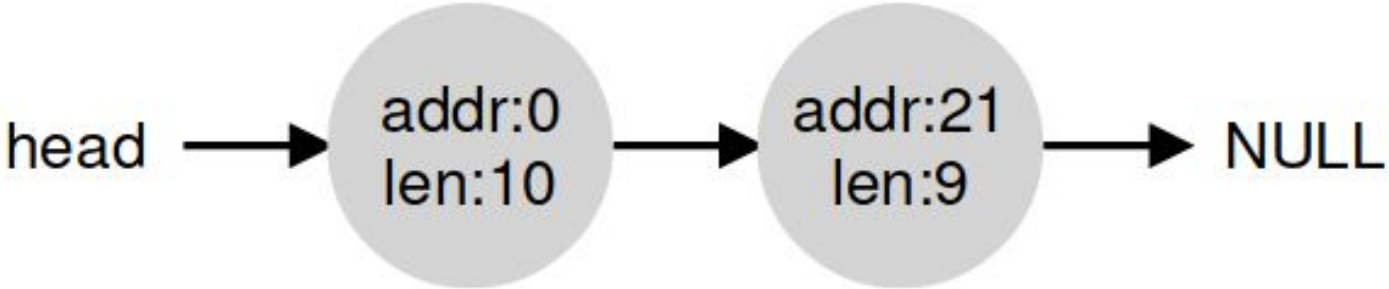
Управление свободным пространством

Лекция №11


free
used
free
0
10
20
30


free
used
free
0
10
20
30


head
addr:0
len:10
addr:20
len:10
NULL


head
addr:0
len:10
addr:21
len:9
NULL


head
addr:10
len:10
addr:0
len:10
addr:20
len:10
NULL


head
addr:0
len:30
NULL

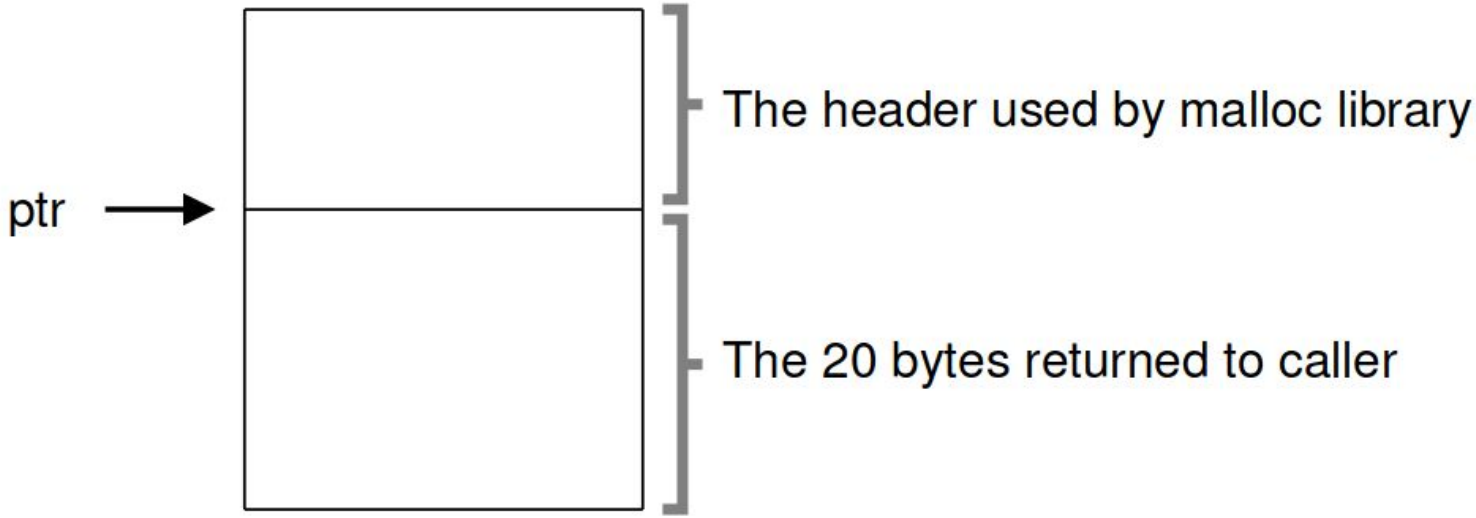


Figure 17.1: **An Allocated Region Plus Header**

```
typedef struct __header_t {
    int size;
    int magic;
} header_t;
```

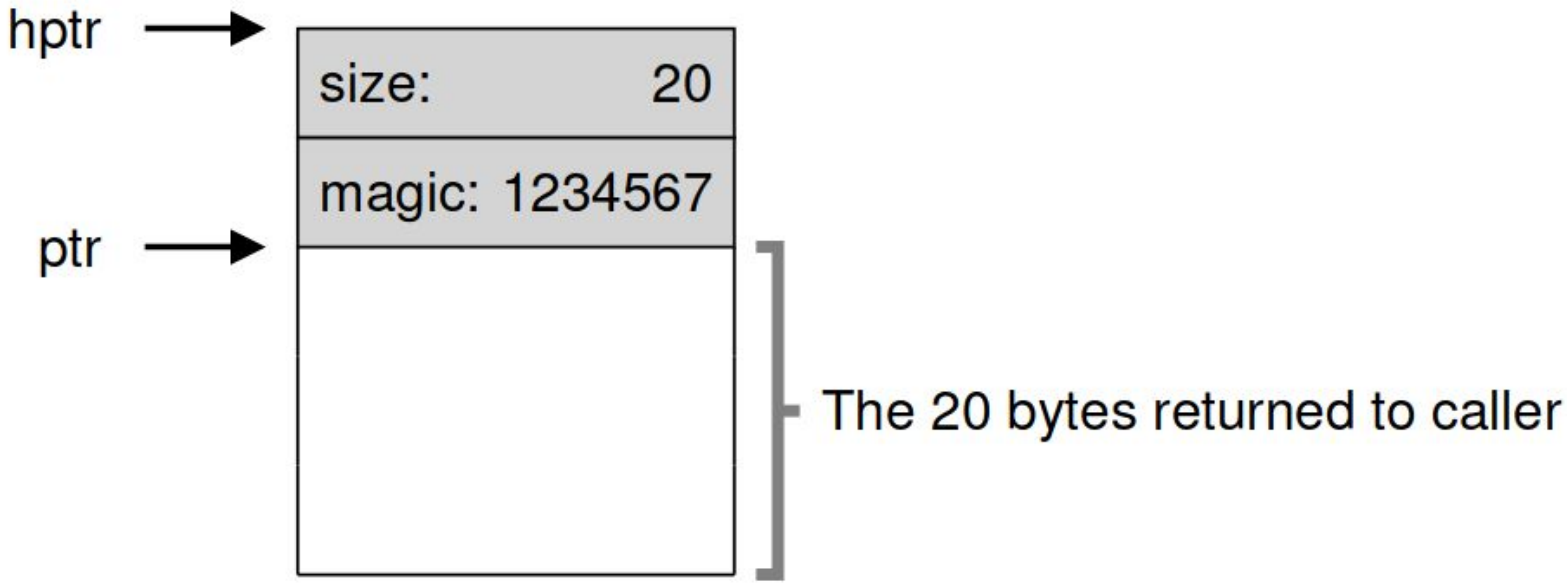


Figure 17.2: **Specific Contents Of The Header**

```
void free(void *ptr) {
    header_t *hptr = (void *)ptr - sizeof(header_t);
    ...
```

```
typedef struct __node_t {
    int                 size;
    struct __node_t *next;
} node_t;
```

```
// mmap() returns a pointer to a chunk of free space
node_t *head = mmap(NULL, 4096, PROT_READ|PROT_WRITE,
                    MAP_ANON|MAP_PRIVATE, -1, 0);
head->size   = 4096 - sizeof(node_t);
head->next   = NULL;
```

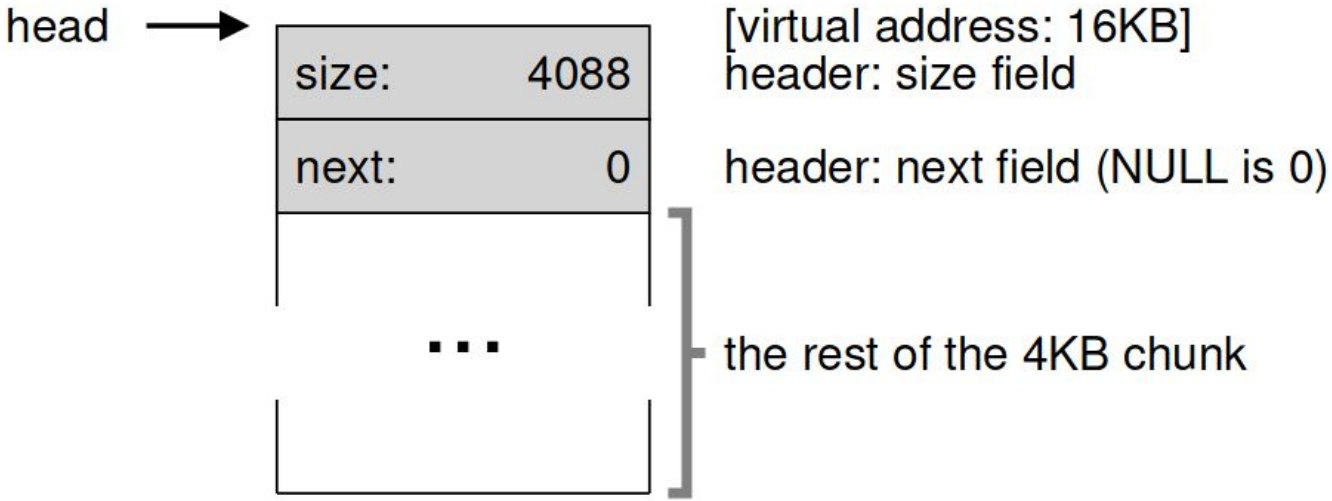


Figure 17.3: **A Heap With One Free Chunk**

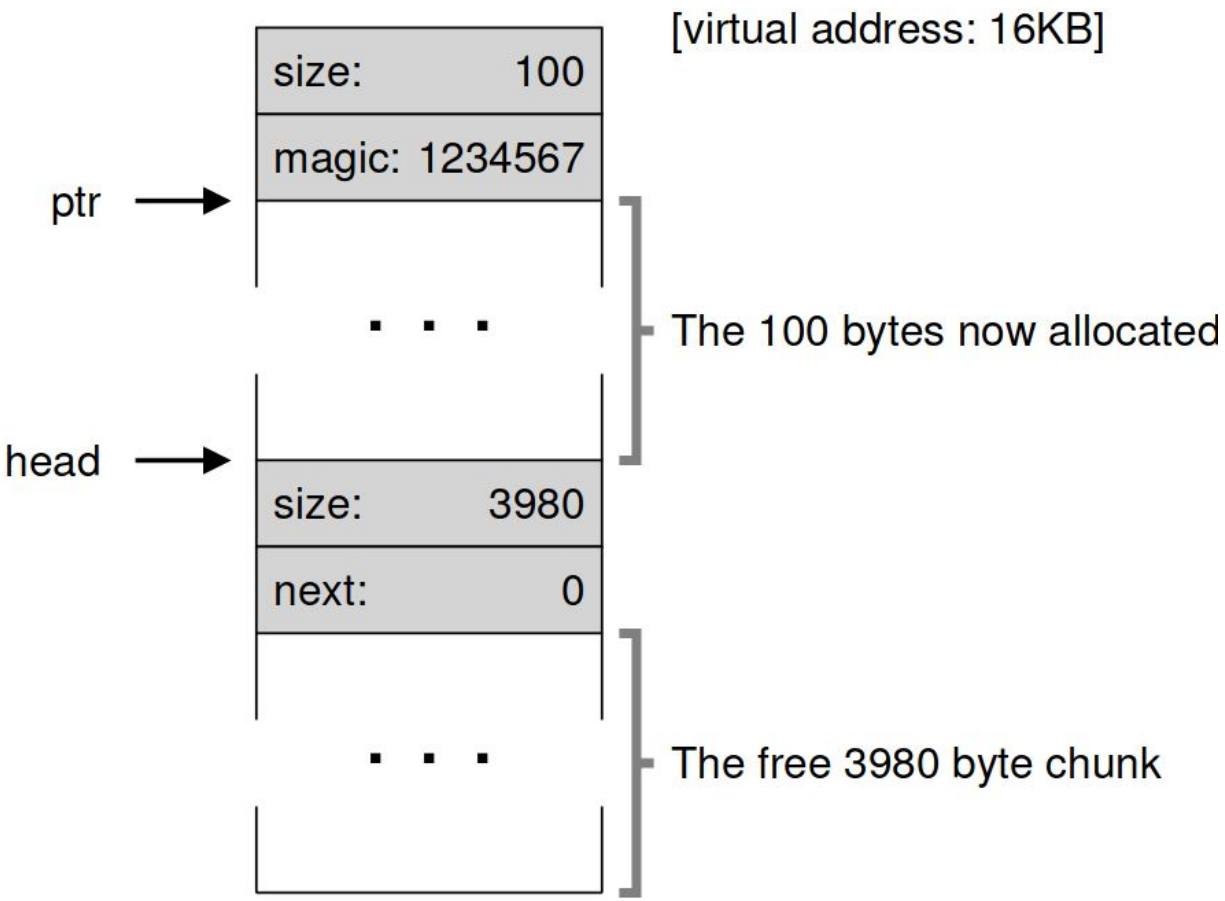


Figure 17.4: **A Heap: After One Allocation**

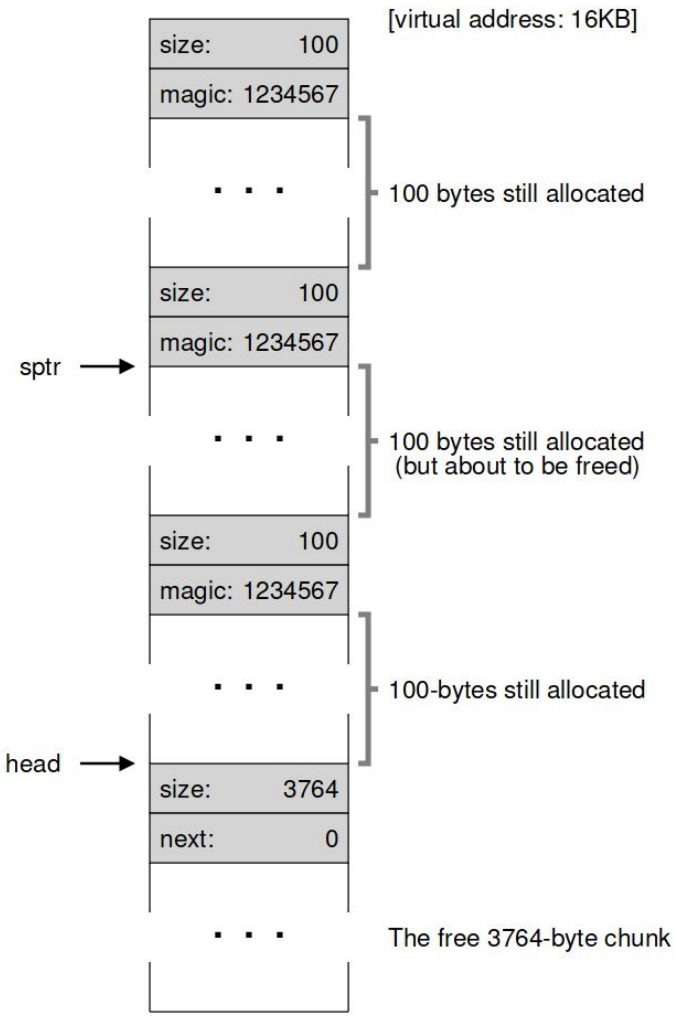


Figure 17.5: **Free Space With Three Chunks Allocated**

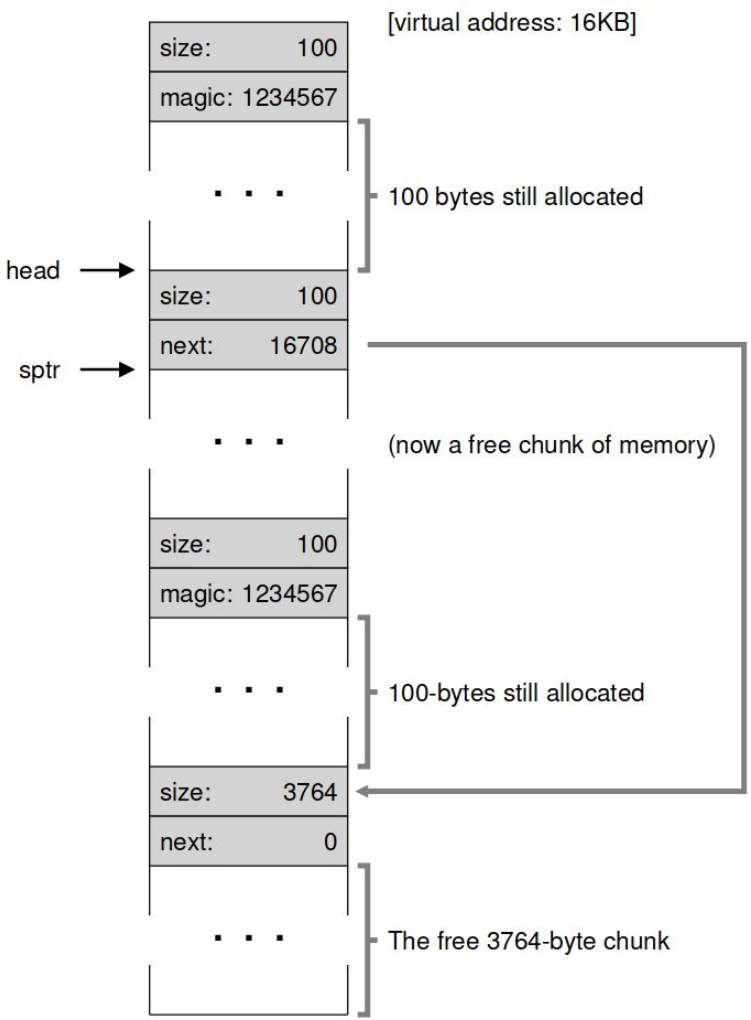


Figure 17.6: **Free Space With Two Chunks Allocated**

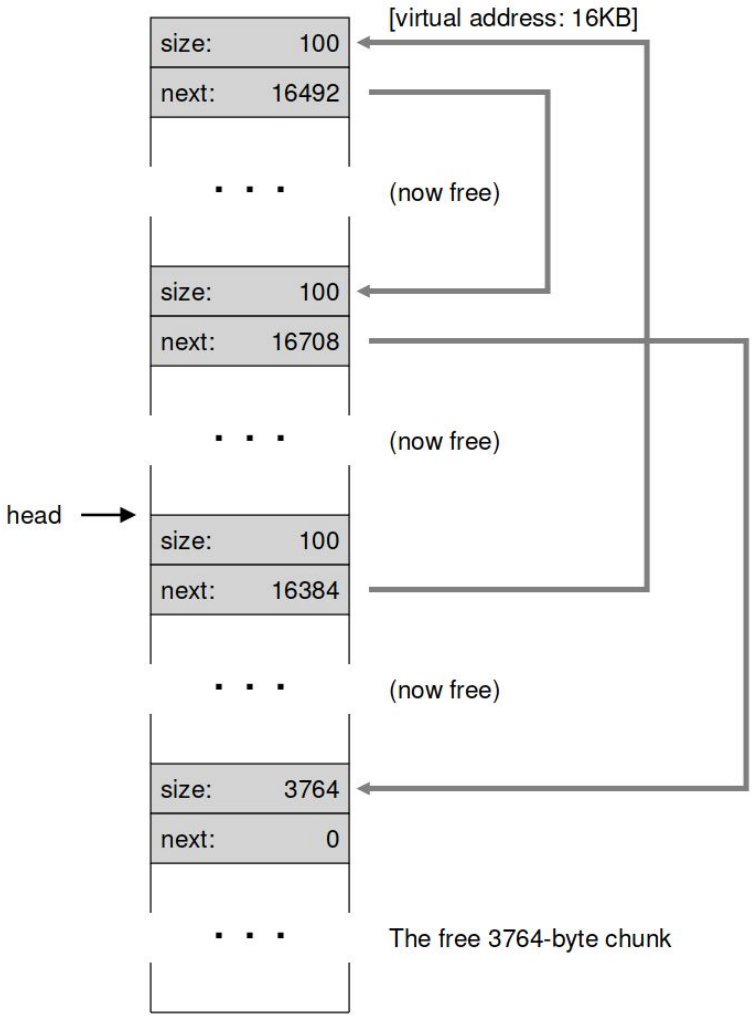
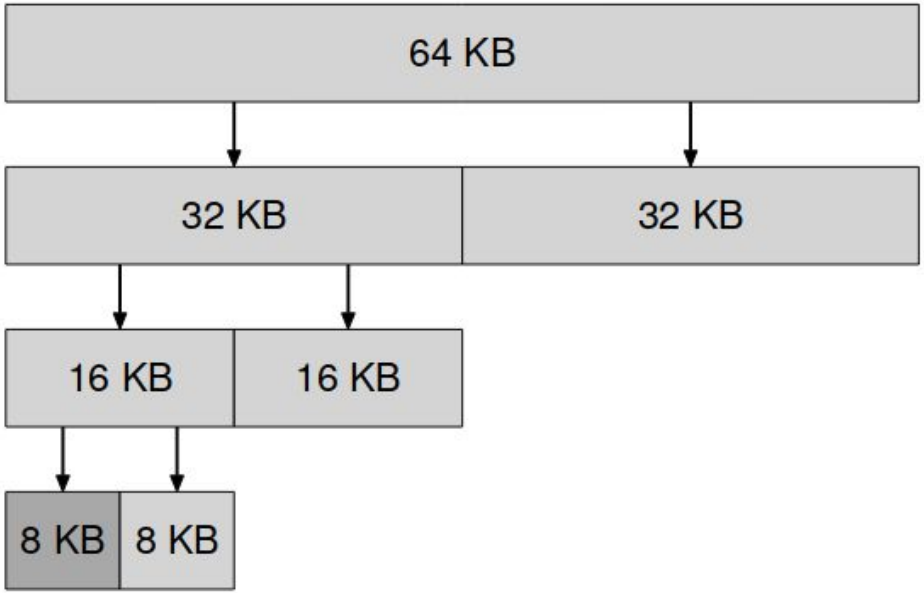


Figure 17.7: **A Non-Coalesced Free List**


head
10
30
20
NULL


head
10
30
5
NULL


head
10
15
20
NULL


64 KB
32 KB
32 KB
16 KB
16 KB
8 KB
8 KB